Practical 4: Atomistic Simulation

Course: Materials Simulation Practical

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Introduction

Atomistic simulations play a crucial role in understanding the microscopic origins of material properties by modeling interactions between atoms using interatomic potentials. The accuracy and transferability of these potentials directly determine how well simulated quantities such as cohesive energy, lattice constants, and elastic moduli reproduce experimental or first-principles data. In metallic systems like copper (Cu), where bonding arises from a delocalized electron cloud, it is essential to employ many-body potentials that account for collective electronic effects rather than simple pairwise interactions. This practical explores the differences between two-body and many-body potential models, focusing particularly on the Effective Medium Theory (EMT) and Embedded Atom Method (EAM), which are designed to capture metallic bonding more realistically.

Many-Body vs Two-Body Interactions:

Effective-Medium Theory (EMT) and the Embedded Atom Method (EAM) are many-body potentials, whereas a simple pair potential like Lennard-Jones (LJ) includes only two-body interactions. In EMT/EAM, the energy per atom depends on the local environment density (electron density or neighbor density) in addition to pairwise distances. For example, EAM assigns each atom an embedding energy that is a function of the host electron density contributed by neighbors, plus a pairwise term. In contrast, a pair potential like LJ sums

energies for each atom pair independent of other neighbors. This means EMT/EAM implicitly include multi-atom effects (bond order or cohesion changing with coordination), whereas pair potentials assume fixed two-atom interaction regardless of environment [1].

Limitations for Metals:

Simple pair potentials like LJ are generally not suitable for metals such as copper because metallic bonding is fundamentally a many-body phenomenon. Metals have a delocalized electron gas; the bond energy of an atom depends on all its neighbors collectively, not just on individual pairs. A pair potential cannot capture effects like the dependence of cohesion on coordination number or the correct energy difference between various crystal structures. For instance, with a pair potential one usually cannot simultaneously fit a metal's cohesive energy, equilibrium lattice parameter, and elastic constants, and the functional form is too restrictive. Furthermore, the pairwise potentials automatically obey certain elastic relations that real metals do not (Cauchy relations), meaning a simple pair potential will predict incorrect relative values of elastic constants for copper. They also fail to predict the correct vacancy formation energies or surface energies for metals, because removing one neighbor in a metal has a non-linear effect on bonding that pair potentials can't reproduce [5].

In essence, metals require potentials (like EMT, EAM, or other multi-body potentials) where bond strength is environment-dependent. The only real advantages of pair potentials are their simplicity and computational efficiency. A pair potential (like LJ or Morse) has very few parameters and a simple functional form, making it easy to implement and fast to compute. Moreover, having fewer parameters means they are easier to fit for those properties they can reproduce (often just equilibrium distance and bonding energy). However, this simplicity comes at the cost of realism for metals [5].

Parameter Fitting: Pair Potentials vs. Semi-Empirical Potentials:

- Pair Potentials (e.g. LJ, Morse): These have very few parameters (two for LJ: σ and ϵ). Fitting is straightforward but limited – typically one chooses parameters to match basic properties such as the equilibrium bond distance (or lattice constant) and cohesive energy of the metal. For example, the LJ well depth ϵ might be set to reproduce copper's cohesive energy and σ to get the correct nearest-neighbor distance. However, once these two are fixed, all other predictions are determined. There is no flexibility to independently tune the bulk modulus, elastic constants, defect energies, etc. As a result, pair potentials cannot simultaneously fit multiple target properties for a metal; any fit is a compromise. They often end up with incorrect elastic moduli or thermal behavior because the functional form is too restrictive. In essence, with pair functions a good fit to one property can force wrong values for others (e.g. a Lennard-Jones potential fitted to Cu's cohesion and lattice will typically give an incorrect bulk modulus and melting point) [2].

- Semi-empirical Many-Body Potentials (EMT/EAM): These models have more parameters and sometimes functional forms that can be adjusted, so fitting them is more involved but also more powerful. Typically, EAM/EMT parameters is fitted to a large database of experimental or ab initio data: not only the equilibrium lattice constant and cohesive energy, but also elastic constants (C_{11}, C_{12}, C_{44}), vacancy formation energies, stacking fault energies, surface energies, etc. This multi-objective fitting is often done with non-linear optimization algorithms to minimize error against many target properties. The

presence of the embedding function and pair function in EAM means there are several functions (often represented by tabulated values or splines with adjustable parameters) that can be tuned. For example, the shape of the electron-density function or embedding energy curve can be adjusted to get the correct bulk modulus without altering the equilibrium lattice parameter. As a result, EAM potentials can be made to reproduce a broad set of a metal's properties fairly well. The downside is the fitting process is complex – there may be dozens of parameters, and fitting requires careful attention to avoid overfitting one property at the expense of others. In practice, published EAM potentials for Cu (such as the Mishin et al. 2001 potential mentioned below) use on the order of 8–10 adjustable parameters and have been optimized to reproduce many properties of copper simultaneously. This yields a more transferable potential [4].

Theory

Lattice constant

In crystalline solids, the lattice constant defines the periodic spacing between atoms in the crystal lattice. It reflects the equilibrium distance that minimizes the potential energy of atomic interactions. For cubic systems such as face-centered cubic (fcc) and body-centered cubic (bcc) metals, this geometric relationship can be directly linked to the metallic radius.

In the figures below, copper (Cu) is used as an example of the face-centered cubic (fcc) structure and α-iron (Fe) as an example of the body-centered cubic (bcc) structure. Atoms are represented as equal spheres and the in-plane unit shapes are overlaid for clarity. For fcc, the closest-packed plane is (111). On this plane, the nearest-neighbor spacing equals the in-plane lattice vector length $d = \frac{\ell}{\sqrt{2}}$. Because neighbors touch in the close-packed plane, $a = d$, which gives the standard fcc relation:

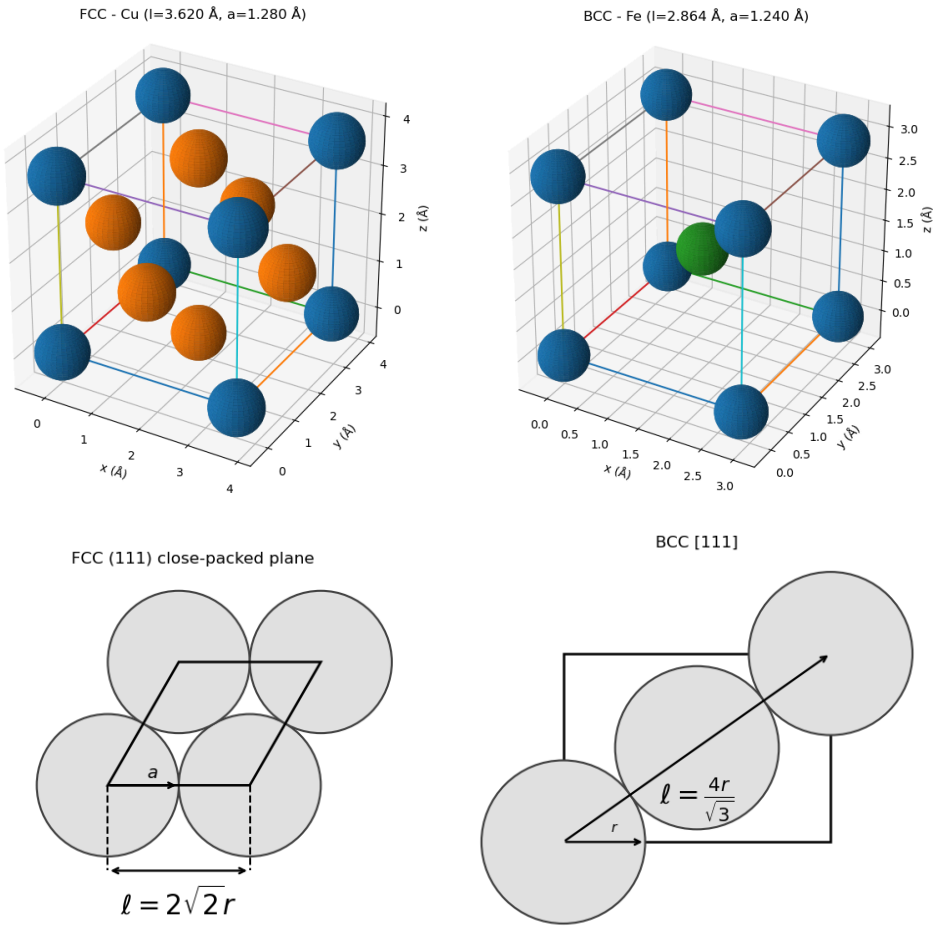
$$\boxed{\ell = 2\sqrt{2}a}$$

(Equivalently, four radii lie along the face diagonal $\sqrt{2}\ell$, again yielding $\ell = 2\sqrt{2}a$.) For bcc, a strictly close-packed plane does not exist, the densest direction is the body diagonal [111]. Along this line, corner–center–corner atoms touch, so the body diagonal $\sqrt{3}\ell$ spans four radii, giving:

$$\boxed{\ell = \frac{4a}{\sqrt{3}}}$$

This contact can be visualized in the (110) cross-section, where the [111] line appears as the rectangle's diagonal. Using these relations with typical metallic radii illustrates the sketches: for Cu with $a \approx 128\ \text{pm},\ \ell \approx 2\sqrt{2} \approx 0.362\ \text{nm}$ for Fe with $a \approx 124\ \text{pm},\ \ell \approx \frac{4a}{\sqrt{3}} \approx 0.286\ \text{nm}.$

Note: These values are the expected hard-sphere estimates, and small deviations from experimental lattice constants are expected due to bonding details and temperature.



Elastic constants

The elastic constants describe how a crystal responds to small deformations and are key parameters linking atomic-scale interactions to macroscopic mechanical behavior. For cubic crystals such as fcc Cu, the stiffness tensor is simplified to a form that depends only on two independent parameters: C_{11} and C_{12}. The shear modulus components ($C_{44} = C_{55} = C_{66}$) are derived from C_{11} and C_{12} as $\frac{C_{11}-C_{12}}{2}$.

In an atomistic framework, these constants are obtained by applying small, prescribed strain tensors to an equilibrium unit cell and computing the resulting changes in potential energy. Assuming purely elastic behavior, the strain energy density W^e is quadratic in the strain $\boldsymbol{\varepsilon}$, which leads to the relation:

$$W^e = \frac{1}{2}\boldsymbol{\varepsilon} : \boldsymbol{C} : \boldsymbol{\varepsilon} = \frac{1}{2}\left(\boldsymbol{\varepsilon}^v\right)^{\mathsf{T}} \boldsymbol{C}^v_{iso} \boldsymbol{\varepsilon}^v$$

By using specific deformation modes (i.e., applying a known strain tensor), the stiffness constants can be determined using energy calculations. To compute C_{11}, a uniaxial strain ($\bar{\varepsilon_u}$) in the 11-direction was applied in the Practical notes:

$$\boxed{C_{11} = \frac{2W^e(\bar{\varepsilon}_u)}{\bar{\varepsilon}_u^2}}$$

To calculate C_{12}, a hydrostatic strain ε_h will be used:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \bar{\varepsilon}_h & 0 & 0 \\ 0 & \bar{\varepsilon}_h & 0 \\ 0 & 0 & \bar{\varepsilon}_h \end{bmatrix} \quad \longrightarrow \quad \boldsymbol{\varepsilon}^v = \begin{bmatrix} \bar{\varepsilon}_h \\ \bar{\varepsilon}_h \\ \bar{\varepsilon}_h \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The strain energy density is:

$$W^e = \frac{1}{2}\left(\boldsymbol{\varepsilon}^v\right)^{\mathsf{T}} C_{iso}^v \boldsymbol{\varepsilon}^v =$$

$$= \frac{1}{2}\begin{bmatrix} \bar{\varepsilon}_h & \bar{\varepsilon}_h & \bar{\varepsilon}_h & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} \end{bmatrix} \begin{bmatrix} \bar{\varepsilon}_h \\ \bar{\varepsilon}_h \\ \bar{\varepsilon}_h \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \bar{\varepsilon}_h^2 \left(\frac{3}{2}C_{11} + 3C_{12}\right)$$

Thus,

Methods

To evaluate the equilibrium properties and elastic behavior of copper (Cu) at 0 K, atomistic simulations were performed using the Effective Medium Theory (EMT) potential implemented in the Atomic Simulation Environment (ASE). The procedure consisted of two main stages: (i) determination of the equilibrium lattice constant, and (ii) extraction of the independent elastic constants from energy–strain relationships.

Determination of the Equilibrium Lattice Constant

Total energies per atom were computed for Cu in simple cubic (sc), body-centered cubic (bcc), and face-centered cubic (fcc) lattices as functions of the cubic lattice constant ℓ. For each structure, trial values of ℓ were varied within ±1 % around the hard-sphere estimates derived in the Theory section, and the total energies were evaluated using the EMT calculator.

The energy per atom $W(\ell)$ was then fitted by a low-order polynomial to obtain the equilibrium lattice constant ℓ_0 corresponding to the energy minimum. Specifically, a quadratic function of the form

$$W = p_0 + o_1\ell + p_2\ell^2$$

was used to describe the energy–lattice curve near the minimum. The optimized ℓ_0 served as the reference parameter for subsequent elastic-constant calculations.

Calculation of Elastic Constants

For a crystal with cubic symmetry, three independent elastic constants exist: C_{11}, C_{12}, and C_{44}. Again the same strain–energy approach as in theory part will be used and the determinations of C_{11} and C_{12} will coincide with the isotropic case, since they are again going to be obtained from uniaxial and hydrostatic loadings, respectively. An therefore their determination steps will be omitted here. Unlike the isotropic case, for a general cubic crystal, C_{44} is not coupled to (C_{11}, C_{12}) and independent and must be obtained from pure shear deformation. Accordingly, to determine the C_{44}, a simple shear in the 1–2 plane is used:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 0 & \frac{\gamma}{2} & 0 \\ \frac{\gamma}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \longrightarrow \quad \boldsymbol{\varepsilon}^v = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \gamma \end{bmatrix}$$

The strain energy density is:

$$W^e = \frac{1}{2}C_{44}\gamma^2$$

Thus:

$$\boxed{C_{44} = \frac{2W^e(\gamma)}{\gamma^2}}$$

With the three loading paths above, the elastic constants of a material with cubic symmetry are obtained from energy–strain data as:

$$\boxed{C_{11} = \frac{2W^e(\bar{\varepsilon}_u)}{\bar{\varepsilon}_u^2}, \quad C_{12} = \frac{W^e(\bar{\varepsilon}_h)}{3\bar{\varepsilon}_h^2} - \frac{C_{11}}{2}, \quad C_{44} = \frac{2W^e(\gamma)}{\gamma^2}}$$

Numerical Procedure

The strain–energy density curves are sampled along selected deformation paths and fitted with low-order polynomials following the procedure described in the handout. The elastic constants are then obtained from the second derivatives of these fits at the minimum.

First, the optimal fcc lattice constant ℓ_0 was re-determined in the same manner using the code for lattice constant determination by fitting the energy–volume curve around the initial guess. The relaxed fcc unit cell was then constructed at this lattice constant. To ensure that the reference structure corresponded to a minimum-energy configuration, an internal relaxation of atomic positions was carried out while the cell shape and volume were kept fixed.

Afterwards, three one-parameter deformation paths were applied to the reference cell:

1. **Uniaxial strain** ($\varepsilon_{11} = \bar{\varepsilon}$),
2. **Hydrostatic strain** ($\varepsilon_{11} = \varepsilon_{22} = \varepsilon_{33} = \bar{\varepsilon}$),
3. **Simple shear** in the xy-plane (γ with $\varepsilon_{12} = \gamma/2$).

For each path, the strain was varied symmetrically in a small window around zero, and the total energy was calculated without further atomic relaxations. The strain energy density was then obtained by dividing the energy difference with respect to the reference structure by the undeformed volume V_0.

The resulting $W(\bar{\varepsilon})$ curves were fitted with cubic polynomials of the form:

$$W(\bar{\varepsilon}) = p_0 + p_1\bar{\varepsilon} + p_2\bar{\varepsilon}^2 + p_3\bar{\varepsilon}^3,$$

and the curvature at the stationary point was evaluated using:

$$\left.\frac{d^2W}{d\bar{\varepsilon}^2}\right|_{\min} = 2p_2 + 6p_3\,\bar{\varepsilon}_{\min}.$$

By invoking the relations for cubic symmetry, the three independent elastic constants were extracted:

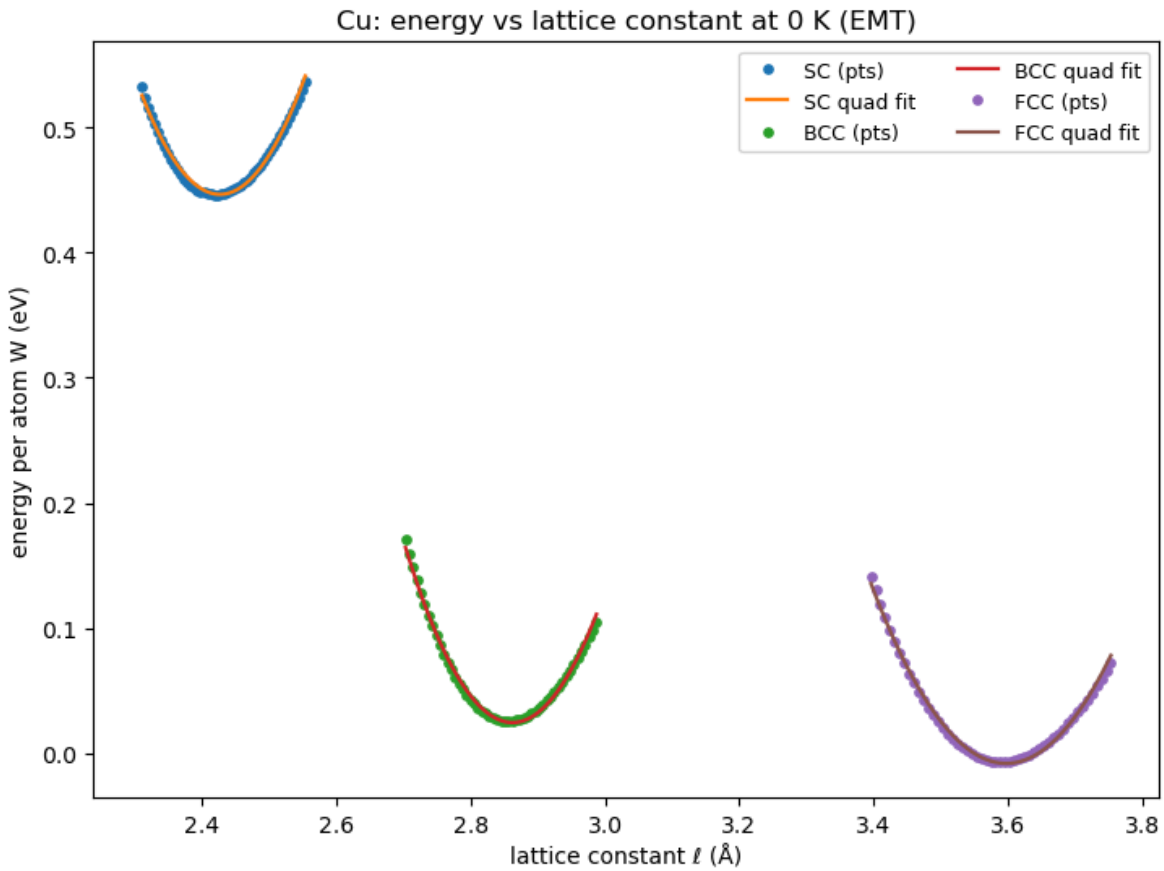
- $C_{11} = \dfrac{d^2W}{d\bar{\varepsilon}^2}$ from the uniaxial path,
- C_{12} from the hydrostatic path via $\frac{d^2W_{\text{hyd}}}{d\bar{\varepsilon}^2} = 3(C_{11} + 2C_{12})$,
- $C_{44} = \dfrac{d^2W}{d\gamma^2}$ from the shear path.

Finally, the values were converted from eV/Å^3 to 10^{11} N/m^2 for comparison with literature

Results

Lattice contants

The equilibrium lattice constant of Cu was obtained by fitting the computed total energy per atom $W(\ell)$ to a quadratic form around the energy minimum. Figure 1 shows the energy–lattice relations for the sc, bcc, and fcc structures evaluated using the EMT potential.



The optimal lattice constant and corresponding minimum energy were extracted from the polynomial fit via:

$$\ell_0 = -\frac{p_1}{2p_2}$$

$$W_0 = W(\ell_0)$$

The fitted values for each structure are summarized as follows:

$\ell_0^{SC} \approx 2.427$ with $W_0^{SC} \approx 0.446$ eV/atom,

$\ell_0^{BCC} \approx 2.862$ with $W_0^{BCC} \approx 0.024$ eV/atom,

and $\ell_0^{FCC} \approx 3.598$ with $W_0^{FCC} \approx -0.008$ eV/atom.

The results yield a stability ordering of

$$W_0^{FCC} < W_0^{BCC} \ll W_0^{SC}$$

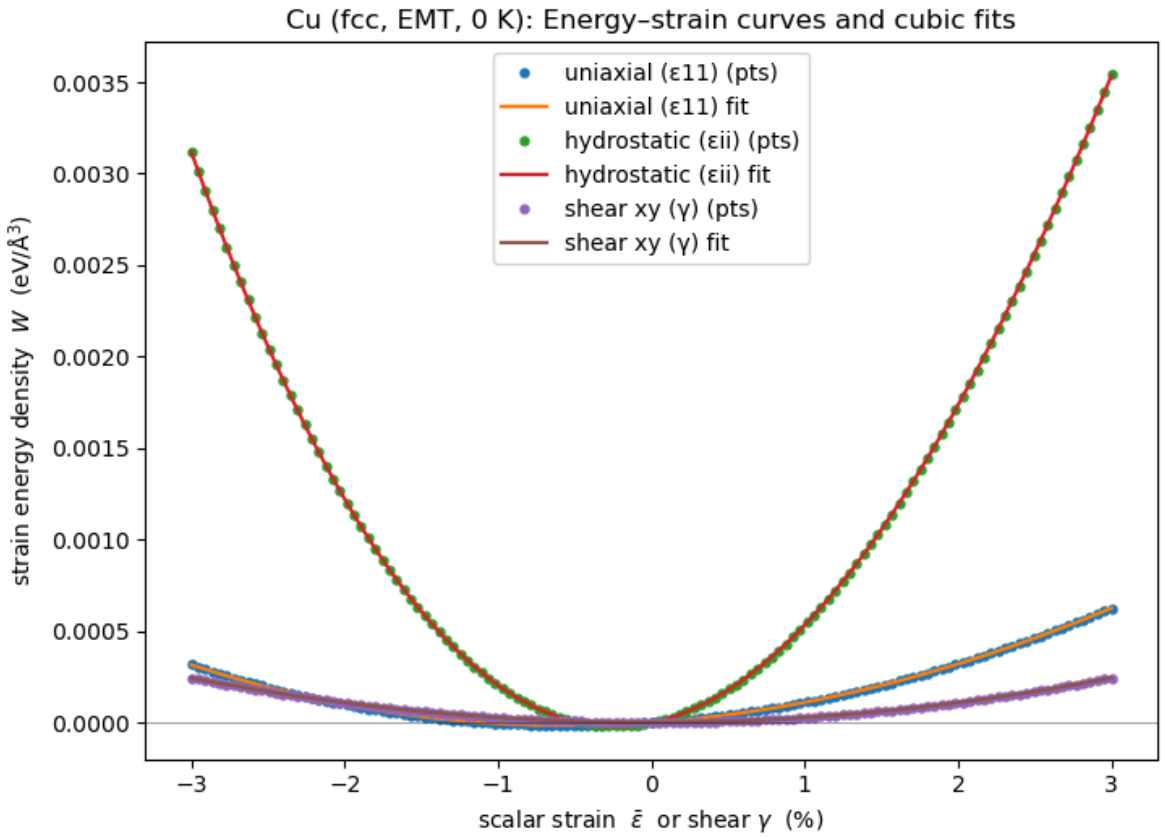
which correctly reproduces Cu's known fcc ground state at low temperature.

The obtained fcc lattice constant deviates from the experimental 0 K value (3.615 Å) by only $\sim -0.48\ \%$, demonstrating good agreement for a semi-empirical potential. It should be noted that the absolute value of W_0 reflects the EMT reference energy and is not a true cohesive energy; for cohesive energy comparisons, the energy must be referenced to an isolated Cu atom.

Elastic constants

To determine the elastic constants, small deformations were applied to the relaxed fcc unit cell following the uniaxial, hydrostatic, and shear strain paths described in the Methods section. The resulting strain–energy density curves were fitted with cubic polynomials, and the elastic constants were extracted from the second derivatives at the stationary points.

Figure 2 illustrates the energy–strain relationships and corresponding polynomial fits.



The obtained elastic constants for fcc Cu at 0 K are summarized below:

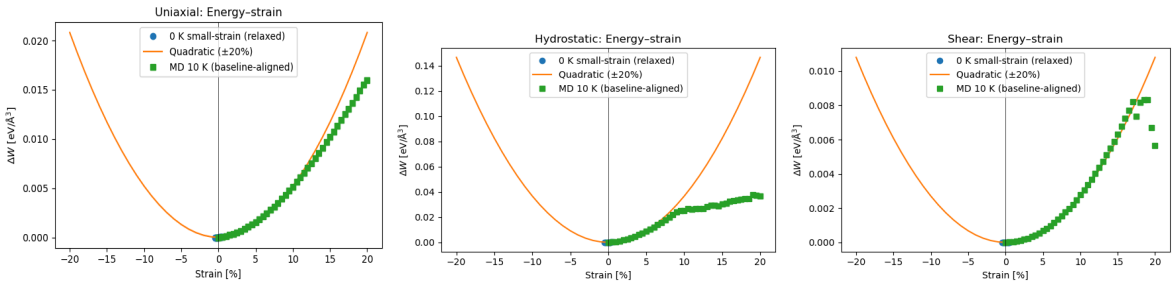
Constant	Value ($\times 10^{11}$ N/m^2)	Ref (0 K)	Error
C_{11}	1.7166	1.7620	-2.58%
C_{12}	1.1617	1.2494	-7.02%
C_{44}	0.8684	0.8177	+6.21%

The EMT potential reproduces the longitudinal stiffness C_{11} with good accuracy, within approximately 2.6% of the reference value. The transverse constant C_{12} is underestimated by about 7%, which is typical for simplified effective medium models. The shear modulus C_{44} is slightly overestimated by ~6%.

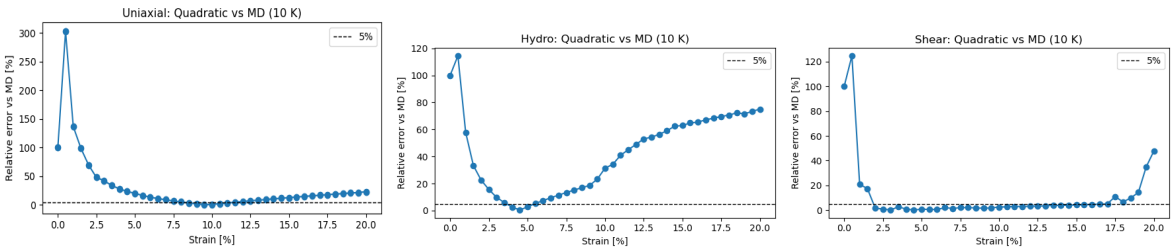
These deviations can be attributed to both the empirical nature of the EMT potential and the numerical fitting procedure. EMT treats metallic cohesion in an averaged, environment-dependent manner but does not explicitly incorporate directional bonding or detailed electronic structure. Consequently, it generally reproduces bulk moduli and equilibrium lattice constants well but tends to deviate more strongly in shear-related quantities, which are sensitive to local bonding anisotropy and electronic rearrangement.

Despite these limitations, the obtained constants remain in physically consistent ratios and capture the correct ordering $C_{11} > C_{12} > C_{44}$, which is characteristic of fcc metals such as copper.

The quadratic strain–energy relation was applied to describe the elastic response of the material near its equilibrium configuration. However, it was necessary to extend and evaluate the range around the relaxed state for which this assumption remains valid. To achieve this, a quadratic fit was obtained from 0 K, position-relaxed structures within ±0.5 % strain and subsequently extrapolated up to ±20 %. The extrapolated curves were then compared with NVT molecular dynamics simulations performed at 10 K, representing quasi-static conditions with minimal thermal fluctuations. Through this comparison, the deviation between the quadratic prediction and the MD results was quantified for uniaxial, hydrostatic, and shear deformations. The resulting plots below illustrate the comparison between the extrapolated quadratic fits and the MD data for each deformation mode.



To more clearly quantify the accuracy of the quadratic approximation across the deformation range, the relative error between the extrapolated fits and the MD results was also evaluated. The following plots display how this error evolves with strain for the three deformation modes.



At very small strains, the computed relative errors show artificially high values due to numerical division by very small energy differences. These deviations are not of physical origin but rather arise from the definition of relative error, which amplifies numerical noise when the reference energies are close to zero. Therefore, the data points in this near-zero strain region were disregarded when determining the validity range of the quadratic approximation.

Based on the resulting plots, the quadratic fit (0 K, ±0.5 %) extrapolated to ±20 % was found to remain within 5 % deviation of the MD data up to **approximately 11.5 % for uniaxial, 5.0 % for hydrostatic, and 16.5 % for shear strain**, beyond which noticeable departures occur due to anharmonic effects.

Carbon Nanotube Chirality and Classification

Carbon nanotubes (CNTs) are cylindrical structures that can be considered as rolled-up sheets of graphene. Their geometry is specified by a pair of integers (n, m), known as the *chirality indices*. These indices define the so-called chiral vector $C_h = n\mathbf{a_1} + m\mathbf{a_2}$, which describes how the graphene lattice is wrapped into a tube. The values of (n, m) determine the nanotube's diameter, its chiral angle θ, and whether the tube belongs to a high-symmetry or a helical configuration [6].

Three main types of CNTs exist:

1. **Zigzag Nanotubes ($m = 0$):**
   Zigzag CNTs are characterized by indices $(n, 0)$. The circumference of the tube follows a continuous zigzag pattern of carbon atoms. Their chiral angle is $\theta = 0^\circ$, meaning the roll-up vector aligns with the graphene's zigzag lattice direction. Zigzag CNTs possess mirror symmetry and are therefore achiral. Their bonds are oriented at approximately 60° relative to the tube axis, which gives them a distinctive bond pattern around the circumference [6].
2. **Armchair Nanotubes ($n = m$):**
   Armchair CNTs correspond to (n, n) indices. The tube circumference resembles the shape of an armchair, with alternating "up-and-down" carbon bonds. The chiral angle is $\theta = 30^\circ$, the maximum possible value, since the roll-up vector lies along the graphene's armchair direction. Armchair tubes also exhibit mirror symmetry and are achiral. In these tubes, some bonds encircle the tube perpendicular to its axis, which influences how they deform mechanically [6].
3. **Chiral Nanotubes ($0 < m < n$):**
   All other (n, m) combinations are classified as chiral. Their atomic arrangement forms a helical pattern around the tube axis, and their chiral angle lies between 0° and 30°. Chiral CNTs lack mirror symmetry and exist in left- and right-handed forms. Their lattice appears as a corkscrew wrapping along the axis, producing unique geometric and mechanical features [6].

Figure 1 below illustrates the roll-up vectors on the graphene lattice that generate zigzag, armchair, and chiral CNTs, and the corresponding atomic arrangements in tube form [7].

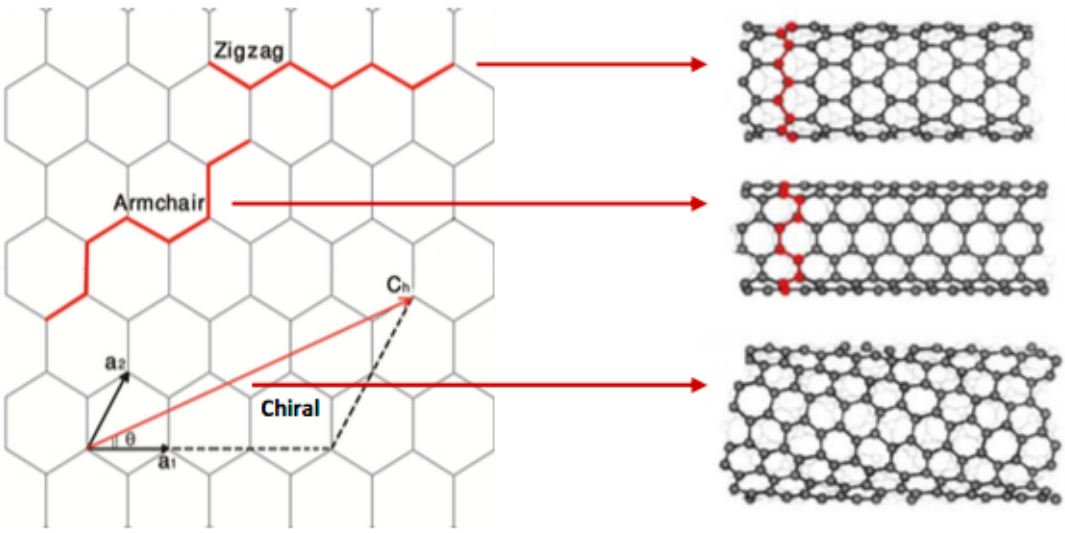


Figure 1. Schematic diagram showing zigzag, armchair and chiral carbon nanotubes.

Although all CNTs share extraordinary stiffness and strength due to strong sp^2 carbon–carbon bonds, differences in atomic arrangement lead to variations in mechanical anisotropy:

- **Axial stiffness and strength:** All types exhibit very high tensile stiffness along the tube axis (modulus on the order of 0.3–1 TPa). Zigzag and armchair tubes generally withstand slightly higher stress compared to chiral tubes, since their symmetric bond orientation allows efficient load transfer [6], [8].
- **Shear and torsion:** Armchair tubes, with bonds encircling the tube, tend to resist torsion more strongly than zigzag tubes. Chiral CNTs display unique *stretch–twist coupling*, where axial loading induces a twist and torsional loading causes axial strain. This effect is absent in the symmetric zigzag and armchair tubes [6], [8].
- **Radial compliance:** All CNTs are highly anisotropic, being extremely stiff axially but relatively compliant radially. Radial deformations such as flattening or buckling require much less force compared to axial stretching [6], [8].

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